

RYAN CHIN JIAN HWA

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Seeking a 6-month / 1-year internship starting mid-August 2026

Education

Universiti Malaya

Oct. 2023 – 2027

Bachelor of Computer Science (Data Science) | CGPA: 3.97

- **Honors:** Dean's List (Every Semester)
- **Scholarship:** Jabatan Perkhidmatan Awam (JPA) LSPM

Ryukoku University, Kyoto

Spring Semester 2026

Japan Studies Program

Relevant Experience

Faculty of Computer Science & Information Technology | Universiti Malaya

Teaching Assistant – Data Structures & Fundamentals of Programming

Oct. 2024 – Jul. 2025

- Conducted lab sessions for 60+ students across two courses, teaching Java-based data structures and introductory programming concepts, along with co-authoring lab test questions and grading assignments.

Projects

LLM Multi-Agent System for Stock Portfolio Construction | Python, Gemini, LangChain, MongoDB

Jan. 2026

- Built an LLM-based multi-agent system (Gemini 2.0 Flash) that automates stock portfolio construction for the Bursa Malaysia market using a 4-agent architecture with Chain-of-Thought prompting and Round Robin debates.
- Engineered a data pipeline scraping financial news and extracting historical prices via yfinance, structuring over 1.1 million records into MongoDB for agent-driven fundamental and sentiment analysis.
- Achieved 100% tool selection accuracy and outperformed the KLCI benchmark in 1-month back-testing. Deployed as an interactive Streamlit web application.

Real-Time Crypto Market Tracker | Kafka, PostgreSQL, dbt, Superset, Docker

Apr. 2026

- Built a real-time pipeline ingesting live trades from Binance's WebSocket API through Kafka (KRaft) into date-partitioned PostgreSQL, using asyncpg's COPY protocol for 5–10x faster batch inserts with backpressure.
- Designed incremental dbt models transforming raw trades into OHLCV candlestick aggregations (1m, 5m, 1h) with a star schema and data quality tests, visualized in Apache Superset dashboards.
- Containerized all 6 services with Docker Compose for single-command deployment and reproducibility.

Analysis of Public Transport Ridership in Klang Valley | Python, Power BI

Apr. 2025

- Won **Best Interactive Dashboard** at UKM Data Challenge 4.0, a national-level data science competition.
- Integrated ridership data from 4 transit operators, engineered temporal features (moving averages, YoY growth), and built predictive models (Random Forest, XGBoost) to forecast ridership trends.
- Designed a multi-page Power BI dashboard with drill-through filters and prescriptive recommendations for transport authorities.

Skills & Certifications

Programming Languages: Python, SQL, Java, Scala, R, TypeScript

Data Engineering: Apache Kafka, Apache Spark, Apache Hadoop, dbt, Docker, Apache Superset

AI & Data Science: LLMs (Gemini), LangChain, Multi-Agent Systems, PyTorch, Scikit-learn, Pandas, NumPy

Databases & Tools: PostgreSQL, MongoDB, MySQL, Redis, BigQuery, Streamlit, Power BI, Tableau, Looker

Certifications: AWS Academy Cloud Foundations (2024), Google Cloud Data Analytics Certificate (2026), Google Advanced Data Analytics Professional Certificate (2026)

Spoken Languages: Fluent in English, Bahasa Malaysia, and Mandarin

Featured Awards

Consolation Prize (Top 10) & Best Interactive Dashboard | UKM Data Challenge 4.0

May 2025

Consolation Prize | Data Challenge 3.0 (UKM Statistics Club)

May 2024

SPM Top Scorer Award | 11A+, Chung Ling High School

2022

Leadership / Extracurricular

Persatuan Komputer Universiti Malaya (PEKOM)

Oct. 2024 – Jan. 2025

Head of Contest Protocol Department – UM Game Jam 2024

- Designed rules, themes, and judging criteria while coordinating a team through collaborative planning and task delegation.